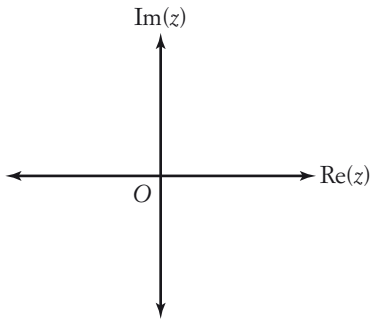
WORKSHEET

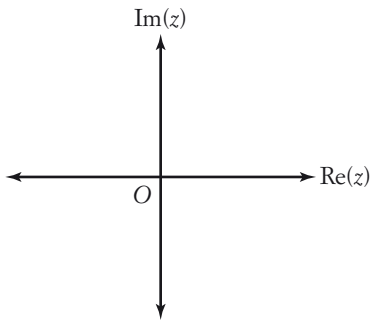
Chapter 4: Division in the plane

1 For each pair of complex numbers z_1 and z_2, find $\left|\frac{z_1}{z_2}\right|$ and $\arg\left(\frac{z_1}{z_2}\right)$. Hence, sketch $\frac{z_1}{z_2}$ on the Argand diagram provided.

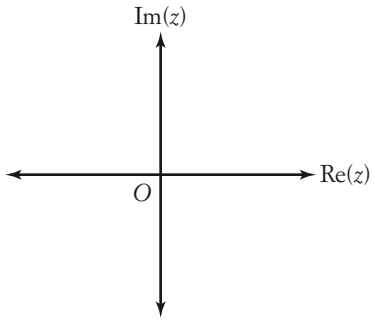
a $z_1 = 12\left[\cos\left(\frac{7\pi}{6}\right) + i\sin\left(\frac{7\pi}{6}\right)\right]$ and $z_2 = 4\left[\cos\left(\frac{\pi}{3}\right) + i\sin\left(\frac{\pi}{3}\right)\right]$



b $z_1 = 2\left[\cos\left(\frac{3\pi}{4}\right) + i\sin\left(\frac{3\pi}{4}\right)\right]$ and $z_2 = 5\left[\cos\left(\frac{\pi}{2}\right) + i\sin\left(\frac{\pi}{2}\right)\right]$

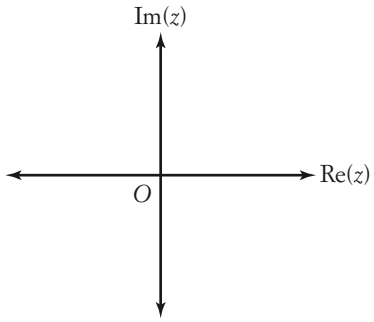


c $z_1 = 7\left[\cos\left(\frac{-2\pi}{3}\right) + i\sin\left(\frac{-2\pi}{3}\right)\right]$ and $z_2 = 5\left[\cos\left(\frac{\pi}{6}\right) + i\sin\left(\frac{\pi}{6}\right)\right]$

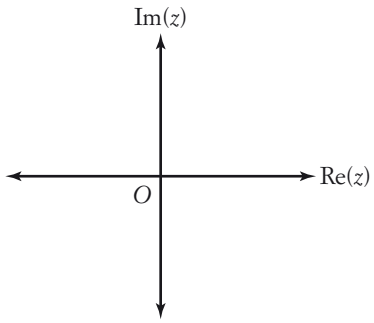


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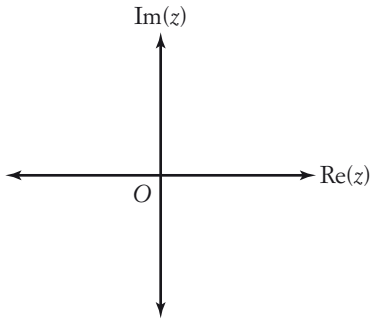
d $z_1 = 3\left[\cos\left(\frac{7\pi}{12}\right) + i\sin\left(\frac{7\pi}{12}\right)\right]$ and $z_2 = 6\left[\cos\left(\frac{-\pi}{4}\right) + i\sin\left(\frac{-\pi}{4}\right)\right]$



e $z_1 = 10\left[\cos\left(\frac{-\pi}{3}\right) + i\sin\left(\frac{-\pi}{3}\right)\right]$ and $z_2 = 8\left[\cos\left(\frac{-2\pi}{3}\right) + i\sin\left(\frac{-2\pi}{3}\right)\right]$



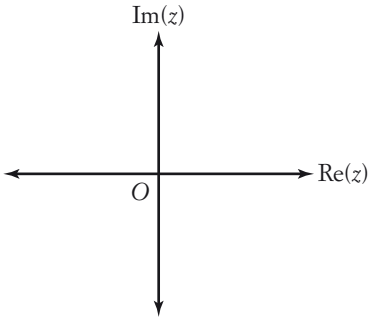
f $z_1 = 6\left[\cos\left(\frac{5\pi}{12}\right) + i\sin\left(\frac{5\pi}{12}\right)\right]$ and $z_2 = 4\left[\cos\left(\frac{\pi}{6}\right) + i\sin\left(\frac{\pi}{6}\right)\right]$



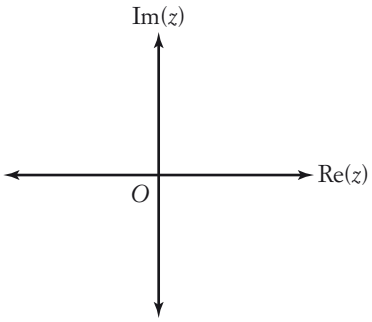
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2 If $z_1 = 3\left[\cos\left(\frac{5\pi}{6}\right) + i\sin\left(\frac{5\pi}{6}\right)\right]$, $z_2 = 5\left[\cos\left(\frac{2\pi}{3}\right) + i\sin\left(\frac{2\pi}{3}\right)\right]$, and $z_3 = 4\left[\cos\left(\frac{\pi}{2}\right) - i\sin\left(\frac{\pi}{2}\right)\right]$ and $z_4 = 6\left[\cos\left(\frac{\pi}{4}\right) + i\sin\left(\frac{\pi}{4}\right)\right]$ sketch each of the following on the Argand plane provided.

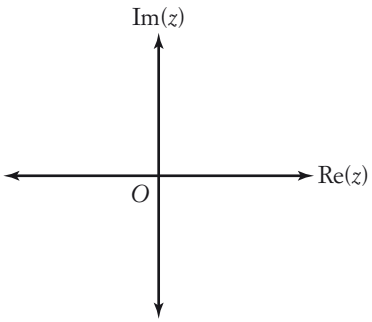
a $\frac{z_1}{z_2}$



b $\frac{z_2}{z_1}$

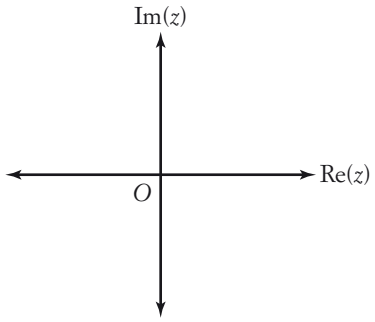


c $\frac{z_3}{z_1}$

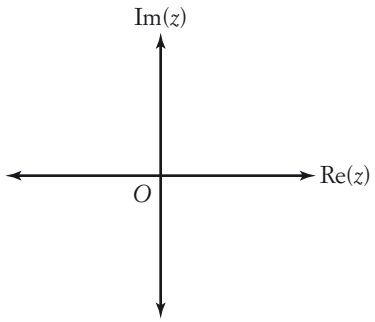


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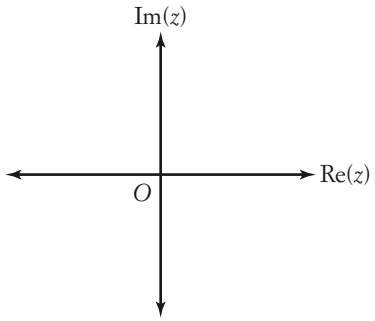
d $\frac{z_2}{z_3}$



e $\frac{z_2}{z_4}$

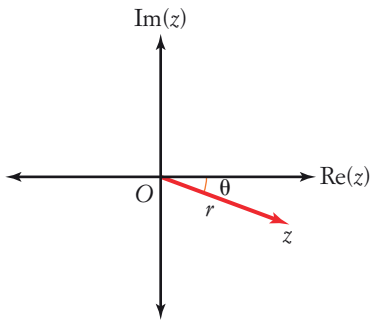


f $\frac{z_4}{z_1}$

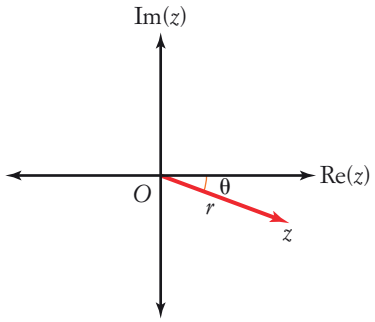


3 Given the vector $\mathbf{z} = (r, \theta)$ representing the complex number z on the complex plane as shown below, sketch each of the following on the diagram supplied in relation to z.

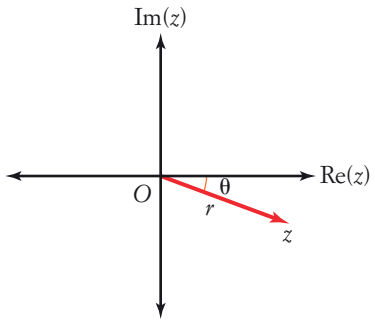
a iz



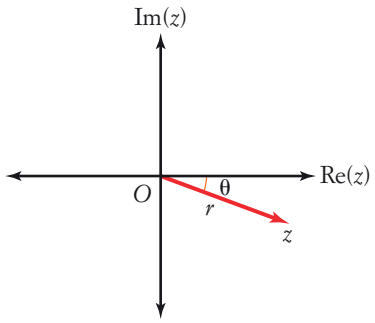
b $\frac{z}{i}$



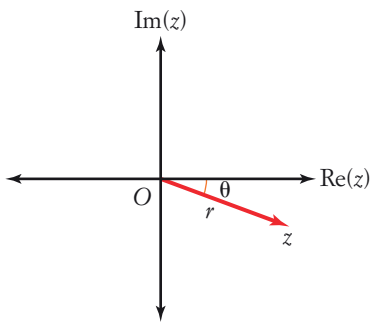
c $-iz$



d $\frac{z}{i^2}$



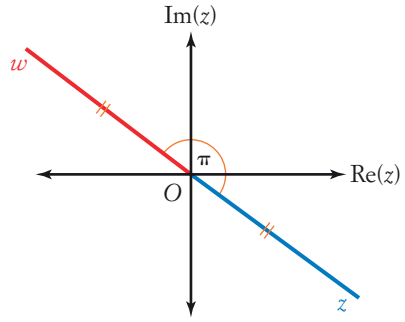
e $-z$



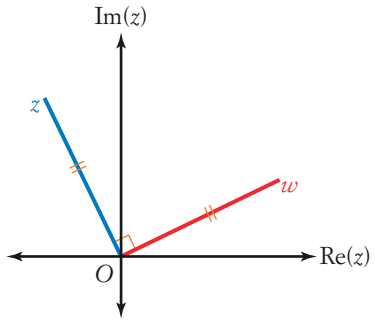
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4 In each case, express the complex number z in terms of w:

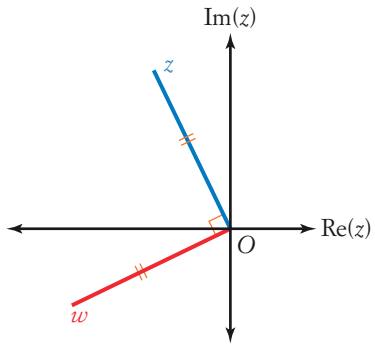
a



b

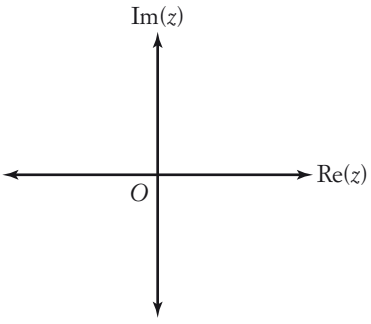


c

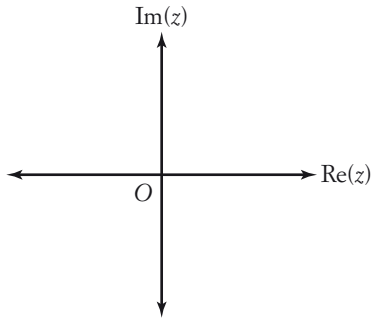


5 Given each complex number z, sketch $\frac{1}{z}$ on the diagram provided:

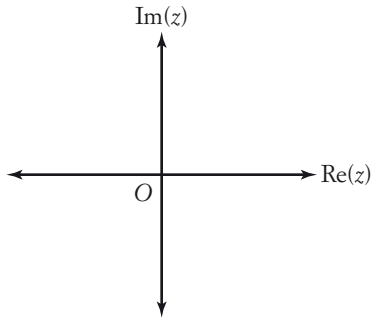
a $z = 3\,\text{cis}\left(\frac{\pi}{4}\right)$



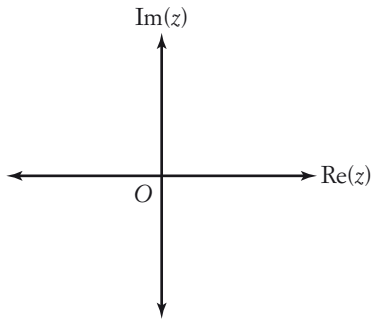
b $z = \sqrt{3}\,\text{cis}\left(\frac{2\pi}{3}\right)$



c $z = 2\sqrt{2}\,\text{cis}\left(\frac{-\pi}{4}\right)$



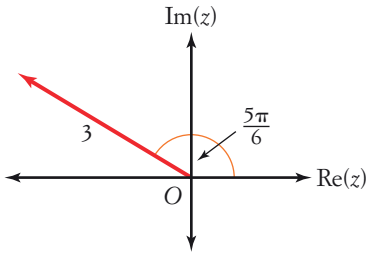
d $z = 2\sqrt{3}\,\text{cis}\left(\frac{-5\pi}{6}\right)$



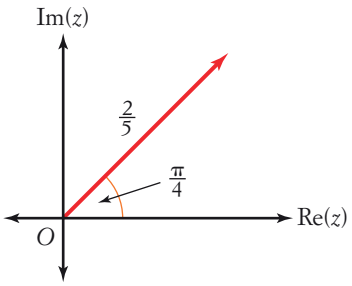
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Answers

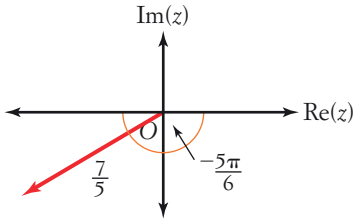
1 a $\left|\frac{z_1}{z_2}\right| = 3, \arg\left(\frac{z_1}{z_2}\right) = \frac{5\pi}{6}$



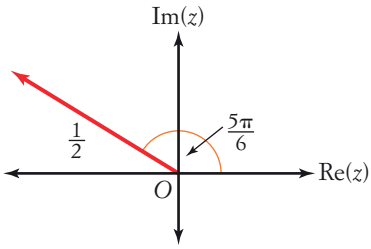
b $\left|\frac{z_1}{z_2}\right| = \frac{2}{5}, \arg\left(\frac{z_1}{z_2}\right) = \frac{\pi}{4}$



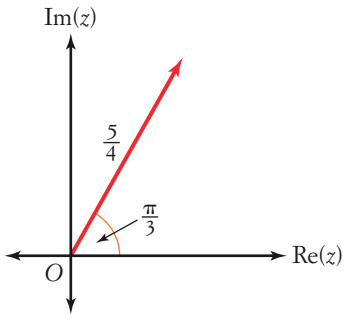
c $\left|\frac{z_1}{z_2}\right| = \frac{7}{5}, \arg\left(\frac{z_1}{z_2}\right) = \frac{-5\pi}{6}$



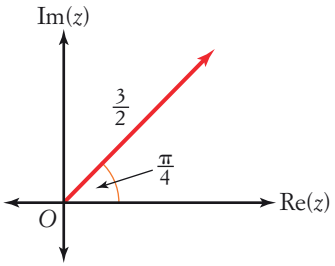
d $\left|\frac{z_1}{z_2}\right| = \frac{1}{2}, \arg\left(\frac{z_1}{z_2}\right) = \frac{5\pi}{6}$



e $\left|\frac{z_1}{z_2}\right| = \frac{5}{4}, \arg\left(\frac{z_1}{z_2}\right) = \frac{\pi}{3}$

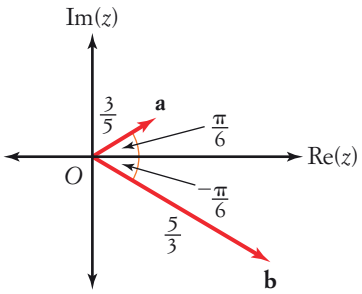


f $\left|\frac{z_1}{z_2}\right| = \frac{3}{2}, \arg\left(\frac{z_1}{z_2}\right) = \frac{\pi}{4}$

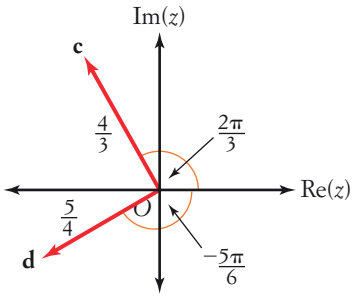


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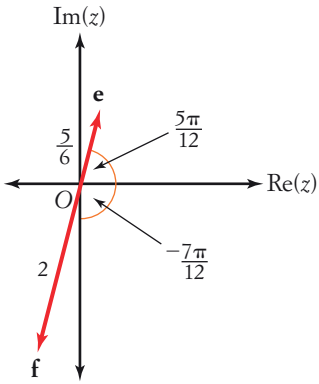
2 **a**



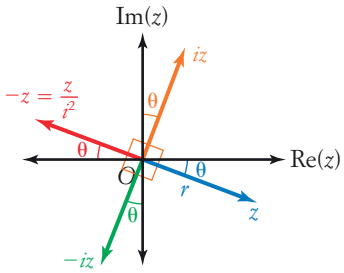
b



c



3

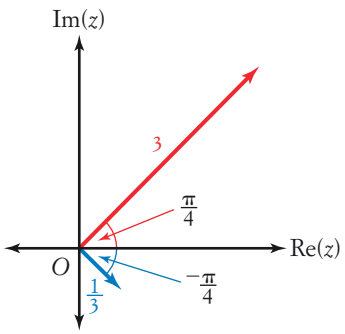


4 **a** $z = -w$

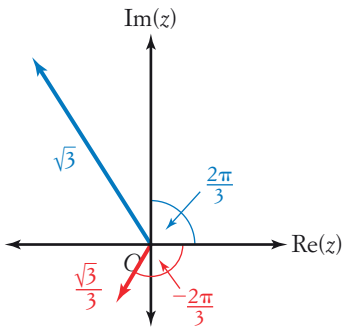
b $z = wi$

c $z = -iw = \dfrac{w}{i}$

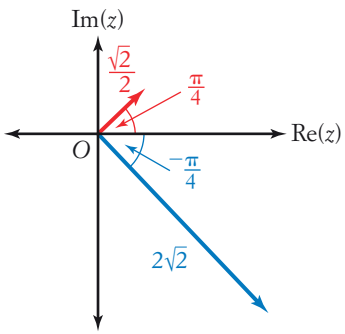
5 **a**



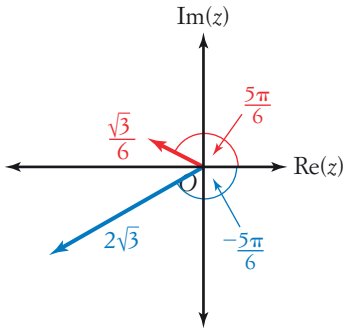
b



c



d



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